

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Data Interpretation

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA0510

Slot A

COURSE OUTCOME COVERED

CO4: Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)

Assessment Tool Weightage: 40%

QUESTIONS:5

Total Marks 50

Duration :60 Minutes Annexure A

Data Interpretation

Code of Conduct:

I AKHILA POCHIMIREDDY(192425225) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Akhila.P

Rubrics for Evaluation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks(100)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data	
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided	
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data	
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions	
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand	

Read the given data carefully and compare the techniques or results based on multiple factors such as accuracy, robustness, and computational constraints. Evaluate and justify your decision with appropriate reasoning considering the application context.

1. Real-Time System Design Trade-off

A surveillance system must operate under constraints:

- Viola-Jones: fast but less accurate (78%)
- YOLO: moderate speed, good accuracy (92%)
- Deep Learning: high accuracy (95%) but slow

Compare the techniques and evaluate which should be selected under strict latency constraints while maintaining acceptable accuracy. Justify your decision considering deployment conditions.

Given Data

- **Viola-Jones:** Fast, accuracy = **78%**
- **YOLO:** Moderate speed, accuracy = **92%**
- **Deep Learning:** Slow, accuracy = **95%**

Answer

Under **strict latency constraints**, **YOLO** is the most suitable technique.

Technique	Speed	Accuracy	Suitability
Viola-Jones	Very Fast	78%	Good for very low latency but lower accuracy
YOLO	Moderate	92%	Best balance
Deep Learning	Slow	95%	High accuracy but unsuitable for strict latency

Analysis:

Viola-Jones has the lowest computational requirement and can process images quickly, but its 78% accuracy may produce more false detections. Deep Learning provides the highest accuracy of 95%, but its slower processing makes it unsuitable when immediate response is required. YOLO provides 92% accuracy with moderate speed, giving a better balance between detection accuracy and processing time.

Conclusion

YOLO should be selected because it provides high accuracy while maintaining acceptable processing speed. It is particularly appropriate for **real-time surveillance**, where both fast response and reliable

2. Robust Detection under Environmental Variations

Detection accuracy across environments:

Method	Normal	Occlusion	Low Light
Viola-Jones	80%	60%	55%
YOLO	90%	75%	70%
Deep Learning	95%	85%	80%

Compare the robustness of each method and evaluate which technique provides the best overall performance across varying conditions. Justify trade-offs.

Given Data

Method	Normal	Occlusion	Low Light
Viola-Jones	80%	60%	55%
YOLO	90%	75%	70%
Deep Learning	95%	85%	80%

The data is provided on page 2 of the assessment.

Analysis

Viola-Jones performs reasonably well in normal conditions at 80%, but its performance decreases considerably under occlusion and low-light conditions.

YOLO performs better, achieving 90% in normal conditions, 75% with occlusion, and 70% in low light. It is therefore more robust than Viola-Jones.

Deep Learning gives the highest accuracy in all three conditions:

- Normal: **95%**
- Occlusion: **85%**
- Low Light: **80%**

The average performance is approximately:

- Viola-Jones = **65%**
- YOLO = **78.3%**
- Deep Learning = **86.7%**

Therefore, Deep Learning has the best overall robustness.

Trade-off

The major disadvantage of Deep Learning is its **higher computational requirement** compared with faster traditional approaches. However, when environmental variations are important, its improved detection reliability is more valuable.

Conclusion

Deep Learning is the best choice for robust object detection across normal, occluded, and low-light environments. It provides the highest overall accuracy and handles environmental changes better than Viola-Jones and YOLO.

3.Optical Flow Reliability vs Noise Sensitivity Two optical flow methods show:

Method A: stable vectors but misses small motions

Method B: detects fine motion but noisy vectors

Compare both methods and evaluate which is more suitable for motion tracking in dynamic scenes. Justify based on application requirements.

Given Data

- **Method A:** Stable vectors but misses small motions.
- **Method B:** Detects fine motion but produces noisy vectors.

Answer

The two methods have different strengths and weaknesses.

Factor	Method A	Method B
Vector stability	High	Low
Small-motion detection	Poor	Excellent
Noise sensitivity	Low	High
Reliability	High	Moderate
Fine motion tracking	Limited	Excellent

Method A produces stable optical-flow vectors, making it suitable when reliable overall motion information is more important. However, it may fail to detect small or subtle movements.

Method B can identify fine movements that Method A misses. However, the noisy vectors can reduce

tracking reliability and may cause false motion information.

Evaluation

For a **dynamic scene involving small and detailed movements**, Method B is more suitable because detecting fine motion is important. However, its noisy output should be reduced using techniques such as:

- Noise filtering
- Smoothing
- Outlier removal
- Temporal consistency checking

For applications requiring highly stable tracking, Method A may be preferable.

Conclusion

For **fine-grained motion tracking in dynamic scenes**, **Method B is preferable**, provided that suitable noise-reduction techniques are applied. This gives a better balance between sensitivity and reliability.

4. Background Modeling Strategy (GMM)

A GMM-based system shows:

- a. High false positives in dynamic background
- b. Missed detections in static background

Compare GMM performance across scenarios and evaluate whether GMM alone is sufficient. Propose and justify improvements.

Given Data

The GMM-based system shows:

- **High false positives in dynamic backgrounds**
- **Missed detections in static backgrounds**

Answer

A **Gaussian Mixture Model (GMM)** represents the background using multiple Gaussian distributions. It is useful for separating moving objects from the background, but its performance depends strongly on the characteristics of the scene.

Performance Analysis

Dynamic Background:

Examples include moving trees, waving leaves, rain, or water. These movements can be incorrectly classified as foreground objects, producing **false positives**.

Static Background:

If the model is not updated appropriately, changes in illumination or object appearance may cause objects to be missed, resulting in **false negatives**.

Therefore, GMM alone is **not sufficient for reliable detection in all environments**.

Proposed Improvements

The system can be improved by combining GMM with:

1. **Adaptive background updating** – continuously update the background model.
2. **Morphological filtering** – remove small noise regions.
3. **Shadow detection** – prevent shadows from being classified as objects.
4. **Object tracking** – maintain object identity across consecutive frames.
5. **Deep-learning-based detection** – improve object recognition in difficult scenes.

Conclusion

GMM alone should not be considered sufficient for complex surveillance environments. A hybrid approach combining GMM with adaptive modeling, filtering, tracking, or deep-learning detection can significantly reduce false positives and missed detections.

4. Motion Estimation vs Computational

Cost Two methods:

- a. Method A: low error (5%), high computation
- b. Method B: moderate error (10%), low computation

Compare the methods and evaluate their suitability for real-time vs offline applications. Justify your decision.

Given Data

- **Method A:** Error = **5%**, high computational cost
- **Method B:** Error = **10%**, low computational cost

Comparison

Factor	Method A	Method B
Error	5%	10%
Accuracy	Higher	Moderate
Computational cost	High	Low
Real-time suitability	Limited	High
Offline suitability	High	Moderate

Real-Time Application

For a **real-time application**, Method B is more suitable because it requires less computation. Although its error is 10%, the lower computational requirement allows faster processing and quicker responses.

Examples include:

- Real-time surveillance
- Traffic monitoring
- Live video analysis
- Robot navigation

Offline Application

For an **offline application**, Method A is preferable because computational time is less critical. Its lower error of 5% provides more accurate motion estimation.

Examples include:

- Video analysis
- Scientific experiments
- Post-event surveillance analysis
- High-accuracy motion measurement

Conclusion

Method B should be selected for real-time applications because its low computational cost enables faster processing. **Method A is better for offline applications** because its 5% error provides greater accuracy when processing time is not a major constraint.

This demonstrates that the best method depends not only on accuracy but also on the **computational requirements and application context**.